Bayesian Hierarchical Logistic Regression for 30-Day Hospital Readmission Among Diabetic Patients: The Role of Medical Specialty

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Abstract

Hospital readmission within 30 days is a costly and clinically important outcome. We investigate whether medical specialty contributes to readmission risk after adjusting for patient-level characteristics, using a Bayesian hierarchical logistic regression model with specialty-specific random intercepts. The model is fitted via a custom Metropolis-within-Gibbs sampler on a cohort of 37,964 diabetic patient encounters across 19 specialty groups. We find substantive between-specialty heterogeneity ($\hat{\tau} = 0.437$, 95% CrI $[0.298, 0.644]$), with Physical Medicine and Rehabilitation showing significantly elevated readmission risk. Key patient-level predictors include prior inpatient visits (OR = 1.22), comorbid neoplasms (OR = 1.51), and A1C measurement (OR = 1.42). Prior sensitivity analysis confirms robustness of all conclusions.

1 Introduction

Unplanned hospital readmissions within 30 days of discharge represent a significant burden on healthcare systems, with estimated annual costs exceeding $26 billion in the United States alone. For patients with diabetes—a population characterised by high comorbidity and complex medication regimens—readmission rates are particularly elevated.

Prior work has identified patient-level predictors of readmission (e.g., length of stay, number of prior admissions, comorbidity burden), typically using standard logistic regression. However, an important structural feature is often ignored: patients are treated by physicians from different medical specialties, and these specialties may differ systematically in their post-discharge care patterns, case mix beyond what is captured by available covariates, or practice style. Ignoring this clustering can lead to biased standard errors and missed opportunities for targeted intervention.

Research question. *Does specialty-level heterogeneity significantly contribute to 30-day readmission risk among diabetic patients, after adjusting for patient-level covariates?*

Why a Bayesian approach? A Bayesian hierarchical model offers three advantages over frequentist alternatives for this problem:

(i) **Partial pooling.** Small-sample specialties (e.g., Physical Medicine, $n = 288$) borrow strength from larger groups through the shared hyperparameter τ, yielding more stable estimates than either complete pooling (ignoring specialty) or no pooling (fixed effects).

(ii) **Coherent uncertainty quantification.** Posterior credible intervals for τ directly quantify our uncertainty about the *magnitude* of between-specialty variation, rather than reducing this to a single p-value.

(iii) **Principled model comparison.** The Deviance Information Criterion (DIC) and posterior predictive checks provide model assessment tools that are native to the Bayesian framework.

2 Data and Cohort Definition

2.1 Data Source

We use the *Diabetes 130-US Hospitals* dataset, which contains 101,766 inpatient encounters from 130 US hospitals over a 10-year period (1999–2008). Each record includes patient demographics, admission details, diagnoses (ICD-9 codes), laboratory results, and medication information.

2.2 Cohort Construction

Starting from $N_{\text{raw}} = 101{,}766$ encounters, we applied the following inclusion criteria:

1. **Remove duplicates and retain first encounter** per patient, yielding one independent observation per individual (removed 30,248 repeated encounters).

2. **Exclude expired/hospice discharges** (removed 1,545 records) to focus on patients at risk for readmission.

3. **Exclude unknown specialty** (removed 32,008 records, 45.7%) since specialty group assignment is central to our research question.

The resulting analytic cohort contains $N = 37{,}964$ encounters across $J = 19$ specialty groups, with an overall 30-day readmission rate of 9.60%.

2.3 Cohort Characteristics

Table 1 summarises the cohort. The mean age is 64.9 years, 53.5% are female, and 73.0% are Caucasian. Readmitted patients tend to be older, have longer hospital stays, more medications, more prior inpatient visits, and more diagnoses on record.

Table 1: Selected cohort characteristics by readmission status. Values are mean (SD) or n (%).

Variable	Overall (N = 37,964)	Readmit = 0	Readmit = 1
Age, years	64.9 (16.3)	64.6 (16.4)	67.9 (14.9)
Length of stay, days	4.3 (3.0)	4.2 (2.9)	4.9 (3.1)
No. lab procedures	42.9 (19.7)	42.8 (19.8)	44.6 (19.2)
No. medications	15.4 (8.4)	15.3 (8.4)	16.6 (8.4)
No. prior inpatient visits	0.2 (0.6)	0.2 (0.6)	0.4 (1.0)
No. diagnoses	6.9 (2.1)	6.9 (2.1)	7.3 (1.9)
Female	53.5%	53.4%	54.6%
Caucasian	73.0%	72.9%	74.1%
Insulin use	51.1%	50.7%	54.5%
Any diabetes medication	76.6%	76.2%	80.5%

Specialty group sizes range from 288 (Physical Medicine and Rehabilitation) to 10,975 (Internal Medicine), with raw readmission rates spanning 4.3% (Obstetrics/Gynecology) to 32.3% (Physical Medicine and Rehabilitation). Figure 1 shows the specialty distribution.

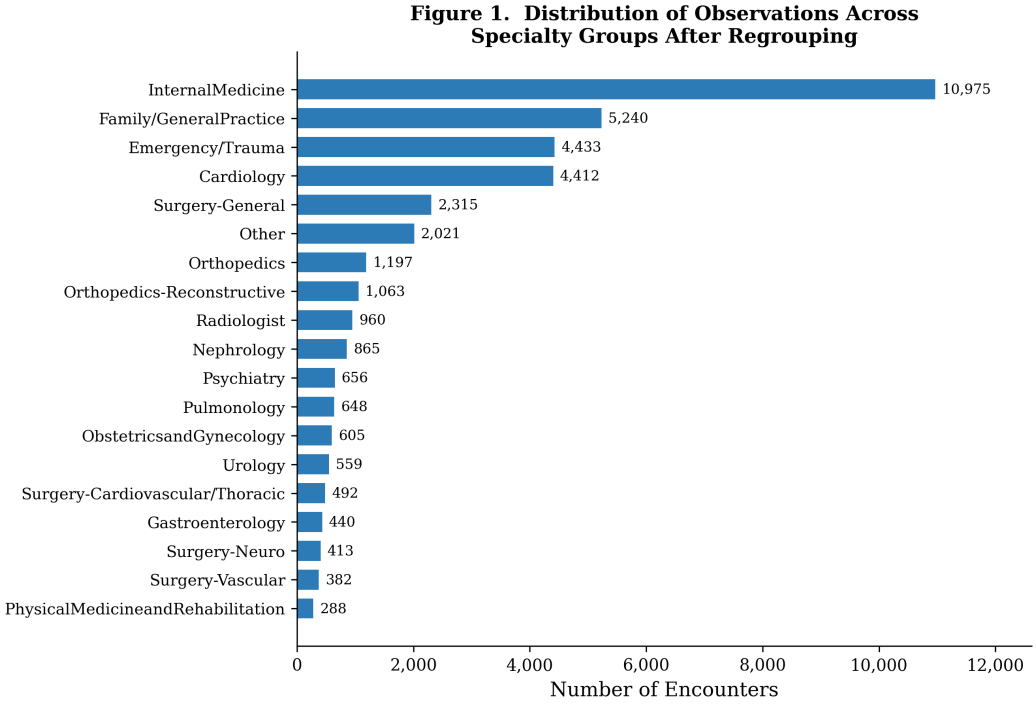


Figure 1: Distribution of encounters across the 19 specialty groups.

3 Model Specification

3.1 Likelihood

Let $y_i \in \{0, 1\}$ denote whether encounter i resulted in 30-day readmission. Conditional on parameters, outcomes are independent Bernoulli:

$$y_i \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u} \overset{\text{ind}}{\sim} \text{Bernoulli}(\pi_i), \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad \eta_i = \alpha + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_{j[i]}, \tag{1}$$

where $\boldsymbol{x}_i \in \mathbb{R}^K$ is the covariate vector ($K = 43$), $j[i] \in \{1, \ldots, J\}$ indicates the specialty group, and $u_{j[i]}$ is the specialty-specific random intercept.

The full likelihood is:

$$p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}) = \prod_{i=1}^{N} \frac{\exp(y_i \, \eta_i)}{1 + \exp(\eta_i)}. \tag{2}$$

3.2 Prior Distributions

$$\alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \qquad \sigma_\alpha = 5, \tag{3}$$

$$\beta_k \overset{\text{iid}}{\sim} \mathcal{N}(0, s_\beta^2), \qquad s_\beta = 2.5, \quad k = 1, \ldots, K, \tag{4}$$

$$u_j \mid \tau \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \qquad j = 1, \ldots, J, \tag{5}$$

$$\tau \sim \text{Half-Normal}(0, s_\tau^2), \qquad s_\tau = 1.0. \tag{6}$$

The priors are weakly informative: $\sigma_\alpha = 5$ places the baseline log-odds within a wide plausible range; $s_\beta = 2.5$ allows odds ratios up to $\sim e^5 \approx 150$; $\tau \sim \text{Half-Normal}(0, 1)$ concentrates mass near zero while permitting substantial heterogeneity.

3.3 Hierarchical Structure

The key feature is the exchangeable random intercepts $u_j \mid \tau \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2)$. The shared scale τ induces *partial pooling*: specialty estimates are shrunk toward the global mean, with the degree of shrinkage inversely related to group size n_j. When $\tau \to 0$, all specialties collapse to a single intercept (complete pooling); as $\tau \to \infty$, each group estimates its own intercept independently (no pooling).

3.4 Assumptions

(i) Conditional independence of outcomes given parameters.

(ii) One observation per patient (first encounter retained).

(iii) Specialty assignment is exogenous (not modelled).

(iv) Only random intercepts; no random slopes.

(v) Prior independence: $\alpha \perp \boldsymbol{\beta} \perp \tau$ *a priori*; $\boldsymbol{u}$ depends on τ only.

4 Posterior Derivation

4.1 Joint Distribution and Posterior

By Bayes' theorem:

$$p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X}) \;\propto\; p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \boldsymbol{X})\, p(\alpha)\, p(\boldsymbol{\beta})\, p(\boldsymbol{u} \mid \tau)\, p(\tau). \tag{7}$$

4.2 Log-Posterior

Substituting the likelihood and priors:

$$\begin{aligned}\ell(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau) = {} & \underbrace{\sum_{i=1}^{N} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big]}_{\text{(I) log-likelihood}} - \underbrace{\frac{\alpha^2}{2\sigma_\alpha^2}}_{\text{(II)}} - \underbrace{\frac{1}{2s_\beta^2} \sum_{k=1}^{K} \beta_k^2}_{\text{(III)}} \\ & \underbrace{- J \log \tau + \frac{1}{2\tau^2} \sum_{j=1}^{J} u_j^2}_{\text{(IV)}} - \underbrace{\frac{\tau^2}{2s_\tau^2}}_{\text{(V)}} + \text{const.}\end{aligned} \tag{8}$$

The logistic likelihood makes this posterior non-conjugate: no parameter block has a closed-form full conditional.

4.3 Full Conditional Distributions

We derive the log-target $h(\cdot)$ for each parameter block by collecting terms from (8) that depend on the parameter being updated.

Block 1: α.

$$h(\alpha) = \sum_{i=1}^{N} \big[y_i \eta_i - \log(1 + e^{\eta_i})\big] - \frac{\alpha^2}{2\sigma_\alpha^2}. \tag{9}$$

Block 2: β_k (for each $k = 1, \ldots, K$).

$$h(\beta_k) = \sum_{i=1}^{N} \big[y_i \eta_i - \log(1 + e^{\eta_i})\big] - \frac{\beta_k^2}{2s_\beta^2}. \tag{10}$$

Block 3: u_j (for each $j = 1, \ldots, J$). Only group-j observations contribute:

$$h(u_j) = \sum_{i:\, j[i]=j} \big[y_i \eta_i - \log(1 + e^{\eta_i})\big] - \frac{u_j^2}{2\tau^2}. \tag{11}$$

Block 4: $\phi = \log \tau$ **(reparametrised).** Applying the change of variable $\tau = e^{\phi}$ with Jacobian $+\phi$:

$$h(\phi) = -J\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^{J} u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi. \tag{12}$$

Importantly, $h(\phi)$ does not involve the likelihood: data inform τ only indirectly through the posterior draws of $\boldsymbol{u}$.

4.4 Posterior Propriety

Proposition 1. *The posterior is proper.*

Proof sketch. The Bernoulli likelihood is bounded ($0 \le p(\boldsymbol{y} \mid \cdot) \le 1$), and all priors ($\alpha$, $\boldsymbol{\beta}$, $\boldsymbol{u} \mid \tau$, τ) are proper distributions. Therefore the joint integrand is bounded above by the product of proper priors, which integrates to 1. □

4.5 Identifiability

The intercept α and random intercepts $\boldsymbol{u}$ are weakly non-identifiable: shifting α by $+c$ and each u_j by $-c$ leaves η_i unchanged. The priors $p(\alpha)$ and $p(\boldsymbol{u} \mid \tau)$ break this symmetry by anchoring both at zero. We further address this computationally with a joint translation move (Block 5 in Section 5).

5 Computational Strategy

5.1 Metropolis-within-Gibbs Sampler

The non-conjugate logistic likelihood rules out Gibbs sampling with closed-form conditionals. Full-dimensional Metropolis–Hastings on the $(1 + K + J + 1) = 64$-dimensional space would have prohibitively low acceptance rates. We therefore use **Metropolis-within-Gibbs (MwG)**, which decomposes the posterior into manageable blocks updated sequentially.

Each block uses a symmetric random walk proposal $\theta^* = \theta^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2_{\text{prop}})$, so the Hastings ratio cancels and the acceptance probability reduces to $\min(1, \exp[h(\theta^*) - h(\theta^{(t)})])$.

5.2 Block Update Schedule

One full iteration proceeds as:

1. **Block 1:** Update α. Cost: $O(N)$ via incremental η update.
2. **Block 2:** For $k = 1, \ldots, 43$: update β_k. Cost: $O(N)$ per coordinate (incremental η shift by $\Delta\beta_k \cdot X_{ik}$).
3. **Block 3:** For $j = 1, \ldots, 19$: update u_j. Cost: $O(n_j)$ per group (only group-j observations evaluated).

4. **Block 4:** Update $\phi = \log \tau$. Cost: $O(J)$.

5. **Block 5:** Joint translation move $(\alpha, \boldsymbol{u})$, repeated $R = 10$ times. Propose $\delta \sim \mathcal{N}(0, \sigma^2_{\text{blk}})$, set $\alpha^* = \alpha + \delta$, $u_j^* = u_j - \delta$. Since η is invariant, the likelihood cancels and only priors appear in the acceptance ratio. Cost: $O(J)$ with no likelihood evaluation.

5.3 Proposal Tuning

Proposal standard deviations are tuned automatically over 10 rounds of pilot runs (2,000 iterations each on a subsample of $n = 20{,}000$). A multiplicative rule scales σ_{prop} by $\times 1.5$ if the acceptance rate exceeds 0.44, and by $\div 1.5$ if below. Per-coordinate proposal SDs are maintained for each β_k and u_j to accommodate varying posterior curvatures.

Final acceptance rates: α: 0.40, $\boldsymbol{\beta}$ (mean): 0.43, $\boldsymbol{u}$ (mean): 0.43, $\log \tau$: 0.37, block: 0.45—all within the recommended 0.20–0.50 range.

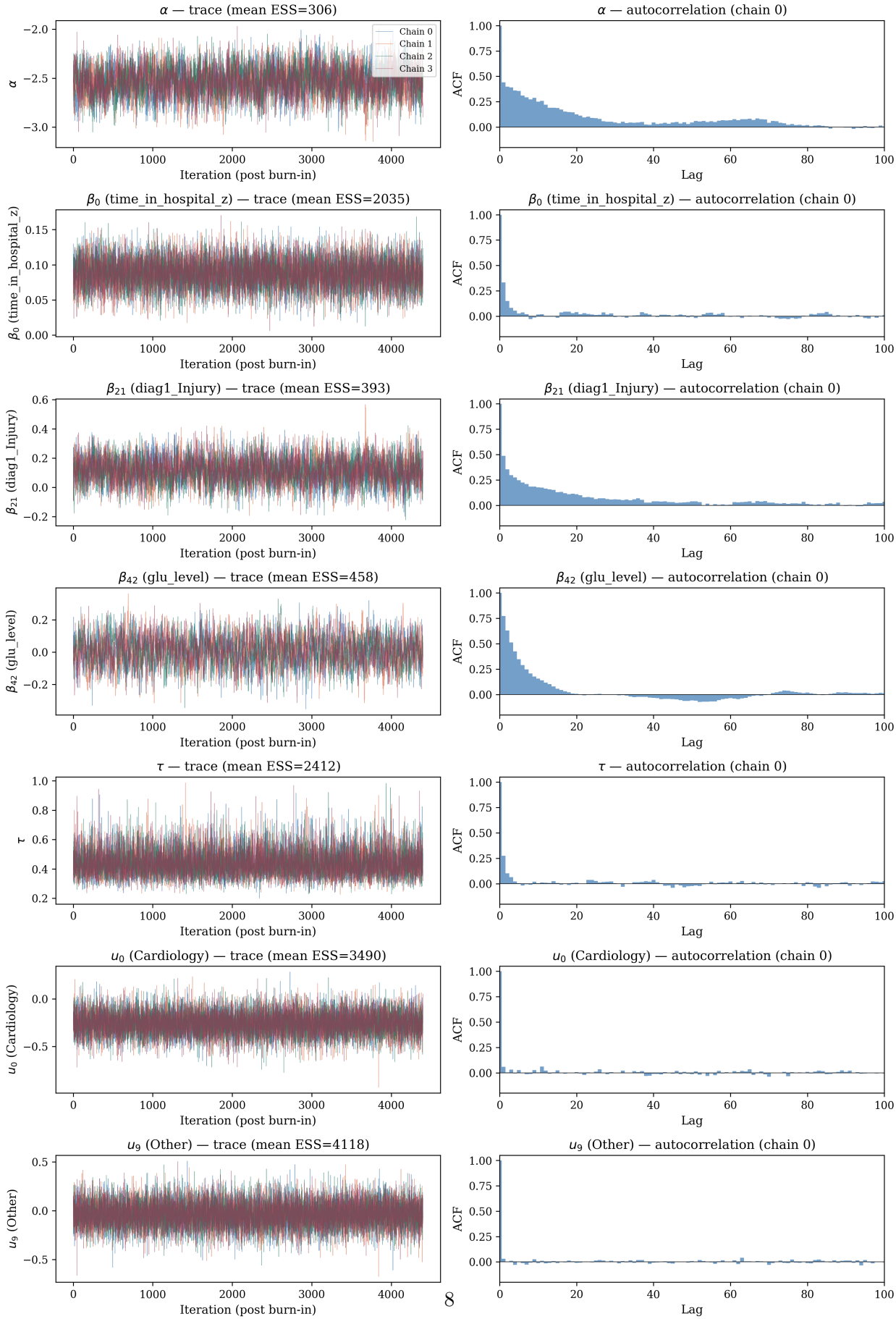
5.4 Chain Configuration

We run 4 independent chains with dispersed random initialisations: 30,000 iterations per chain, burn-in of 8,000, thinning every 5th sample, yielding 4,400 post-burn-in samples per chain and 17,600 pooled samples.

6 Diagnostic Analysis

6.1 Trace Plots and Autocorrelation

Figure 2 shows trace plots and autocorrelation functions for representative parameters. All four chains mix well and overlap, indicating convergence. Autocorrelation decays rapidly to near zero within 20–50 lags for most parameters.


α — trace (mean ESS=306)
Chain 0
Chain 1
Chain 2
Chain 3
α
Iteration (post burn-in)
α — autocorrelation (chain 0)
ACF
Lag
β0 (time_in_hospital_z) — trace (mean ESS=2035)
β0 (time_in_hospital_z)
β0 (time_in_hospital_z) — autocorrelation (chain 0)
β21 (diag1_Injury) — trace (mean ESS=393)
β21 (diag1_Injury)
β21 (diag1_Injury) — autocorrelation (chain 0)
β42 (glu_level) — trace (mean ESS=458)
β42 (glu_level)
β42 (glu_level) — autocorrelation (chain 0)
τ — trace (mean ESS=2412)
τ
τ — autocorrelation (chain 0)
u0 (Cardiology) — trace (mean ESS=3490)
u0 (Cardiology)
u0 (Cardiology) — autocorrelation (chain 0)
u9 (Other) — trace (mean ESS=4118)
u9 (Other)
u9 (Other) — autocorrelation (chain 0)
u18 (Urology) — trace (mean ESS=3245)
u18 (Urology) — autocorrelation (chain 0)

6.2 Effective Sample Size

Table 2 summarises the effective sample size (ESS) across parameter types. All parameters have mean ESS > 100; most exceed 500. The between-group scale τ has ESS $\approx$ 2,412, indicating excellent mixing for the key hyperparameter.

Table 2: Effective sample size summary by parameter type.

Parameter block	Min ESS	Median ESS	Max ESS
α (intercept)	245	306	372
$\boldsymbol{\beta}$ (43 fixed effects)	118	866	3,358
τ (between-group SD)	2,286	2,412	2,661
$\boldsymbol{u}$ (19 random intercepts)	1,593	3,490	4,118

7 Results and Inference

7.1 Fixed Effects

Table 3 presents the posterior summaries for the most important fixed effects. We focus on coefficients whose 95% credible interval (CrI) excludes zero.

Table 3: Key fixed effects: posterior mean, 95% credible interval, and odds ratio. Only coefficients with CrI excluding zero are shown.

Variable	Post. mean	95% CrI	OR	OR 95% CrI
Increased risk				
No. prior inpatient (z)	0.201	[0.175, 0.227]	1.22	[1.19, 1.25]
Comorbid neoplasms	0.415	[0.198, 0.631]	1.51	[1.22, 1.88]
A1C measured	0.352	[0.154, 0.552]	1.42	[1.17, 1.74]
Diabetes medication	0.168	[0.057, 0.277]	1.18	[1.06, 1.32]
Age (z)	0.165	[0.123, 0.208]	1.18	[1.13, 1.23]
No. diagnoses (z)	0.154	[0.111, 0.199]	1.17	[1.12, 1.22]
Comorbid diabetes	0.113	[0.004, 0.224]	1.12	[1.00, 1.25]
Length of stay (z)	0.088	[0.047, 0.128]	1.09	[1.05, 1.14]
No. medications (z)	0.071	[0.024, 0.117]	1.07	[1.02, 1.12]
Decreased risk				
Diag: Neoplasms	−0.507	[−0.792, −0.217]	0.60	[0.45, 0.81]
Diag: Respiratory	−0.360	[−0.519, −0.200]	0.70	[0.59, 0.82]
Diag: Other	−0.347	[−0.515, −0.181]	0.71	[0.60, 0.83]
No. procedures (z)	−0.048	[−0.091, −0.006]	0.95	[0.91, 0.99]

Interpretation. The strongest patient-level predictor of readmission is **prior inpatient visits** (OR = 1.22 per SD increase), reflecting a pattern of recurrent hospitalisation. **Comorbid neoplasms** carries the highest odds ratio (OR = 1.51), suggesting that cancer as a secondary diagnosis substantially elevates readmission risk. Conversely, having **neoplasms as the primary diagno-**

sis is protective (OR = 0.60), likely because these patients are discharged to oncology follow-up pathways.

The A1C measurement indicator (OR = 1.42) likely reflects disease severity rather than a causal effect of testing: sicker patients are more likely to have A1C measured. Notably, sex, race, and insulin use are *not* significant predictors after adjustment.

7.2 Specialty-Level Heterogeneity

Between-group scale τ. The posterior mean of τ is 0.437 (SD = 0.090, 95% CrI [0.298, 0.644]), with $P(\tau > 0.1 \mid \boldsymbol{y}) = 1.0$ and $P(\tau > 0.2 \mid \boldsymbol{y}) = 1.0$. This provides strong evidence that specialty-level heterogeneity persists after adjusting for patient covariates.

Specialty random effects. Figure 3 displays the posterior random intercepts with 95% CrI. **Physical Medicine and Rehabilitation** (PMR) has a markedly elevated random intercept ($\hat{u} = +1.32$, 95% CrI [0.99, 1.67]), the only specialty whose interval is entirely above zero. At the other extreme, **Surgery-Cardiovascular/Thoracic** has a significantly negative effect ($\hat{u} = -0.58$, 95% CrI [-0.99, -0.21]). Most other specialties have credible intervals that include zero, indicating that after covariate adjustment, their readmission rates are consistent with the overall average.

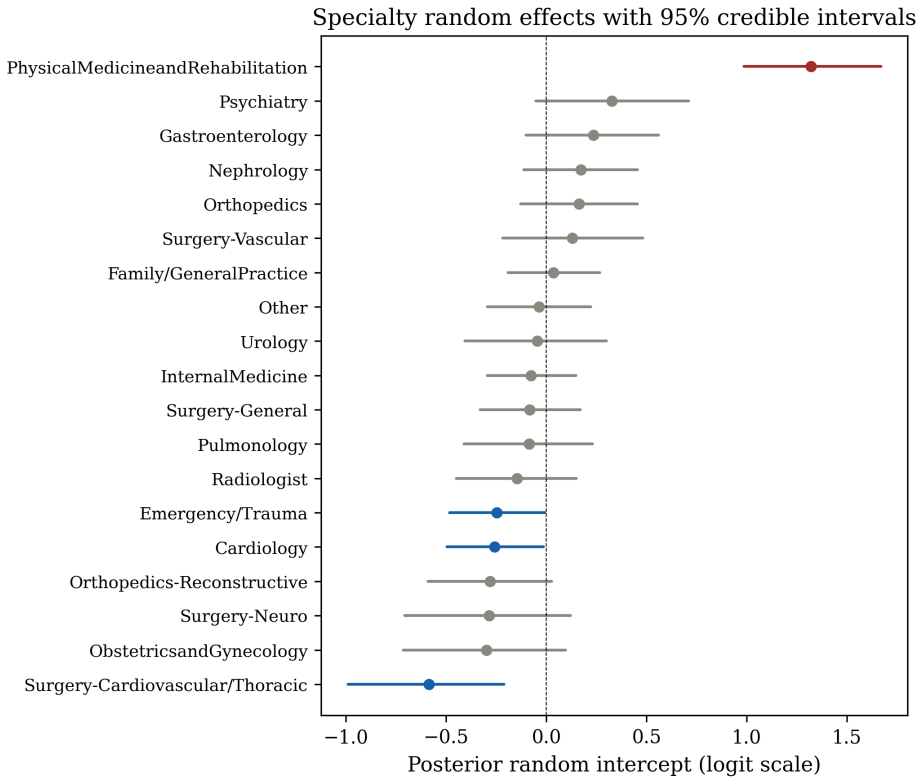


Figure 3: Specialty random effects (posterior mean and 95% CrI). Red: interval entirely above zero; blue: entirely below zero; grey: includes zero.

7.3 Model Comparison

Table 4 compares three models on in-sample predictive performance.

Table 4: Model comparison metrics.

Model	AUC	Brier Score	Log-Loss	Calibration Slope
No specialty (baseline)	0.629	0.0852	0.307	1.003
Specialty fixed effects	0.641	0.0846	0.305	0.997
Bayesian hierarchical	0.641	0.0846	0.305	1.024

The hierarchical model matches fixed-effects performance on discrimination (AUC) and calibration, while offering the advantages of partial pooling for small groups and a coherent estimate of between-group variability.

DIC. The DIC for the hierarchical model is DIC = 23,254 with effective number of parameters $p_D = 60.5$. The nominal parameter count is 64 $(1 + 43 + 19 + 1)$; $p_D < 64$ confirms that the hierarchical prior induces shrinkage on the random intercepts, as expected.

7.4 Posterior Predictive Checks

Figure 4 shows the posterior predictive checks. The observed overall readmission rate (9.60%) falls squarely within the replicated distribution (Bayesian p-value = 0.500), and the observed specialty-rate variability is also well-captured (p-value = 0.245). Both checks confirm adequate model fit.

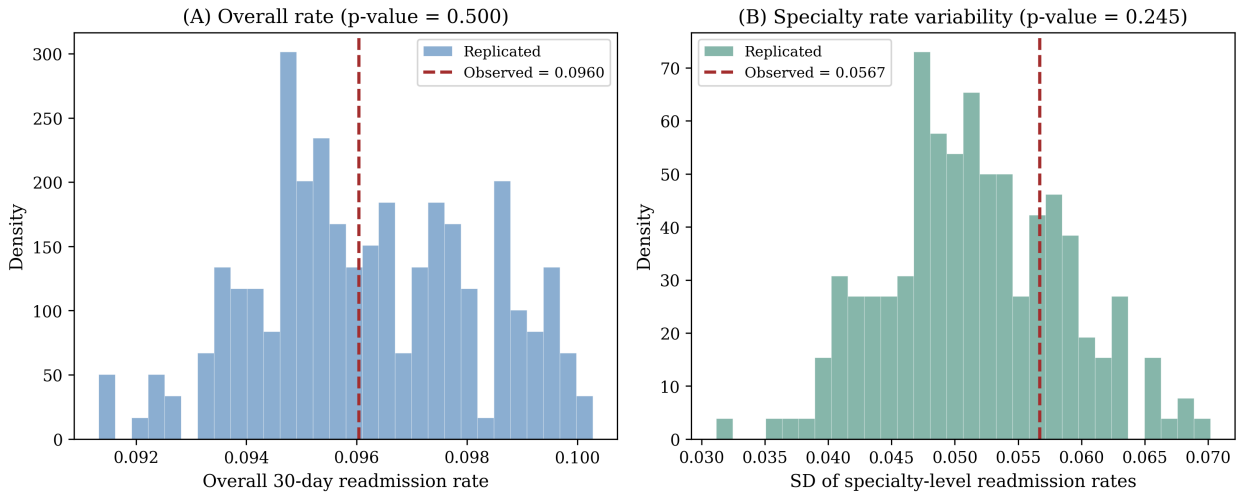


Figure 4: Posterior predictive checks. (A) Overall readmission rate. (B) Standard deviation of specialty-level rates. Dashed red lines show observed values.

8 Sensitivity Analysis

We assess robustness to prior specification by re-running the sampler under three alternative configurations:

- **A1:** Wider fixed-effect prior ($s_\beta = 5.0$).
- **B1:** Wider hyperprior ($s_\tau = 2.5$, Half-Normal).
- **B2:** Heavy-tailed hyperprior ($s_\tau = 2.5$, Half-Cauchy).

Results. Figure 5 shows that the posterior of τ is virtually identical across all four configurations (means: 0.437, 0.437, 0.441, 0.440), confirming that the data overwhelm the prior for this hyperparameter. Figure 6 demonstrates that key fixed effects (e.g., diag1_Neoplasms, comorbid_neoplasms, A1C_measured) are stable across prior specifications.

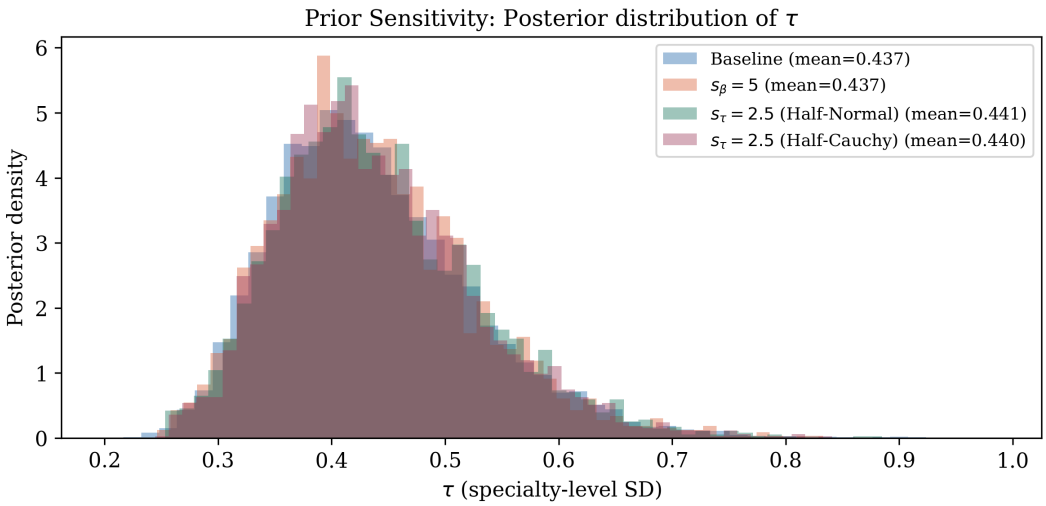


Figure 5: Posterior distribution of τ under four prior configurations.

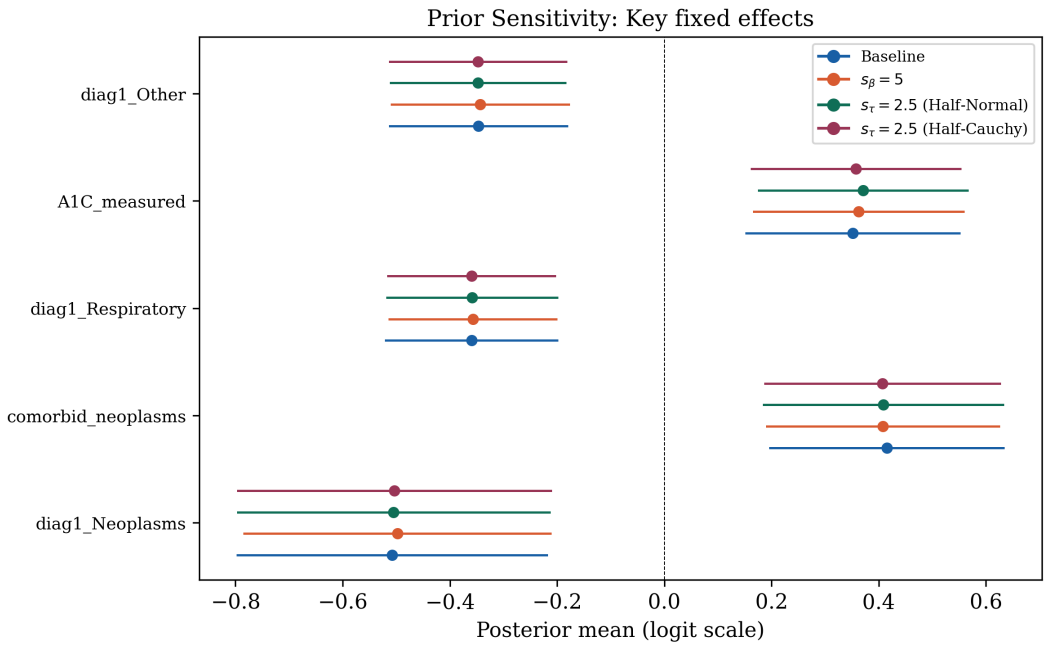


Figure 6: Posterior means and 95% CrI for key fixed effects under four prior configurations.

Conclusion. All substantive findings are robust to reasonable variation in prior specification. This is consistent with the large sample size ($N = 37{,}964$) providing strong likelihood information that dominates the weakly informative priors.

9 Discussion

9.1 Summary of Findings

Our analysis provides three main findings:

1. **Specialty matters.** The between-specialty standard deviation $\tau = 0.437$ (95% CrI [0.298, 0.644]) demonstrates that medical specialty contributes meaningfully to readmission risk beyond what is explained by patient characteristics.

2. **PMR is an outlier.** Physical Medicine and Rehabilitation shows dramatically elevated readmission risk ($u = +1.32$, corresponding to a raw rate of 32.3%), even after adjusting for covariates. This may reflect the unique patient population (e.g., stroke rehabilitation patients with high medical complexity) and could represent a target for quality improvement.

3. **Key patient-level risk factors.** Prior inpatient visits, comorbid neoplasms, A1C measurement, age, and number of diagnoses are the strongest predictors, collectively painting a picture of *medical complexity* as the primary driver of readmission.

9.2 Likelihood-Driven vs. Prior-Driven Behavior

With $N = 37{,}964$ observations, the posteriors of fixed effects $\boldsymbol{\beta}$ are overwhelmingly data-driven: the sensitivity analysis confirms negligible prior influence. The random intercepts exhibit a gradient: large groups (e.g., Internal Medicine, $n = 10{,}975$) are likelihood-driven, while small groups (e.g., PMR, $n = 288$) experience meaningful shrinkage toward zero. The hyperparameter τ is estimated primarily from the empirical spread of the u_j's, with $J = 19$ groups providing sufficient information.

9.3 Limitations

1. **Unknown specialty exclusion.** We dropped 45.7% of encounters with unrecorded specialty. These patients may differ systematically from those with known specialty assignment, potentially introducing selection bias.

2. **No random slopes.** Our model assumes that covariate effects are constant across specialties. In practice, the effect of (e.g.) comorbidity burden may vary by specialty; random slopes could capture this but would substantially increase model complexity.

3. **Temporal and institutional confounding.** The data span 10 years across 130 hospitals, but we do not model hospital-level or temporal effects. Specialty effects may partly reflect unmeasured hospital-level variation.

4. **Observational design.** Specialty assignment is not random; patients are referred based on clinical need. The estimated specialty effects reflect residual confounding as well as genuine practice differences.

5. **Moderate discriminative performance.** AUC = 0.641 reflects the inherent difficulty of predicting readmission from administrative data alone. Clinical notes and social determinants of health—unavailable in this dataset—would likely improve prediction.

9.4 Future Directions

Extensions could include: (i) a crossed random effects model with both specialty and hospital groupings; (ii) random slopes for key covariates; (iii) a time-varying model to capture temporal trends in readmission; (iv) incorporation of the "unknown specialty" group via a mixture model.

Code Availability

All code is available at `https://github.com/Seliiiiina/BayesHLR-Readmission`. The analysis pipeline consists of seven Python scripts (`src/01` through `src/07`), each corresponding to a stage of the analysis. The MCMC sampler is implemented from scratch without external Bayesian modelling software.